Machine Learning for DNA Artifact Classification
Low Template DNA (LTDNA) Analysis

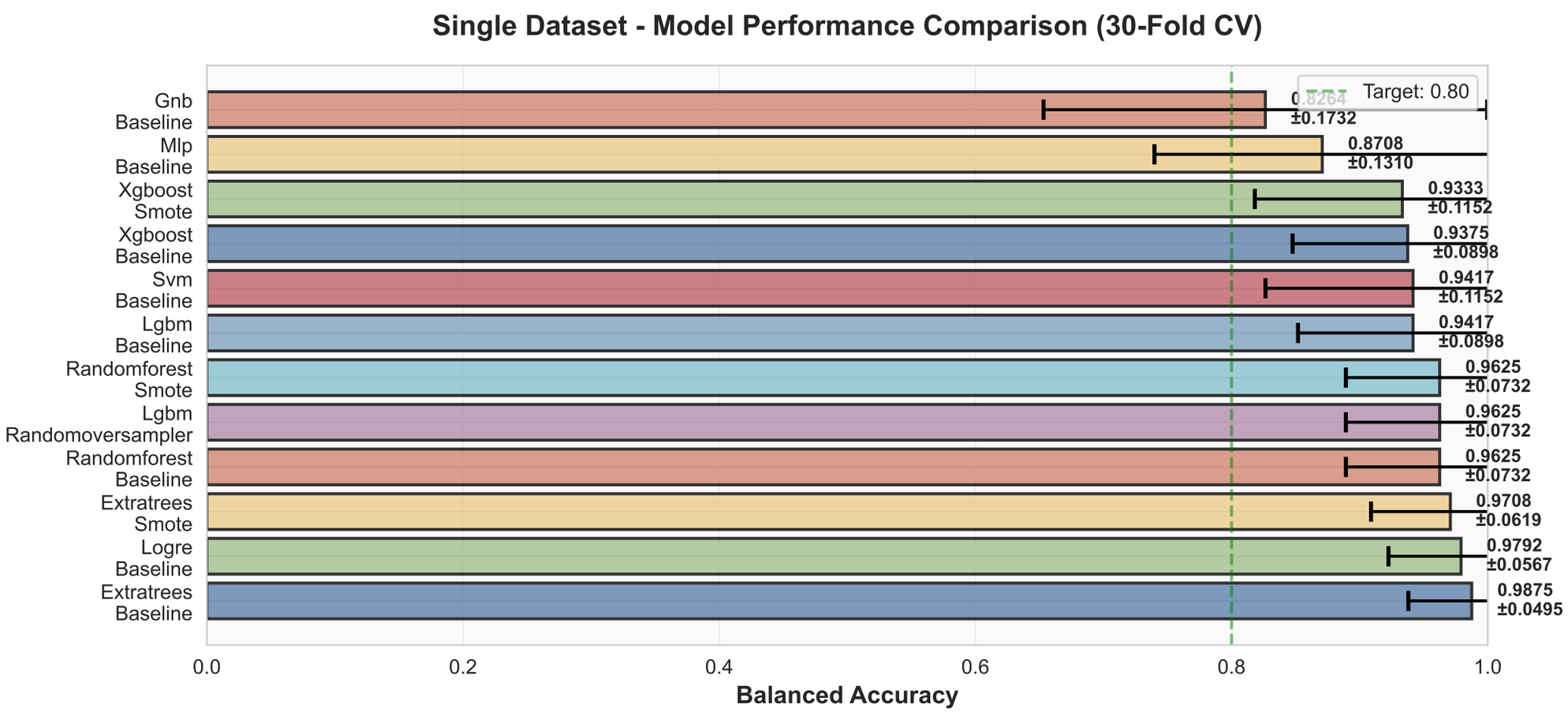
EXECUTIVE SUMMARY REPORT
Cross-Validation Results (30-Fold)

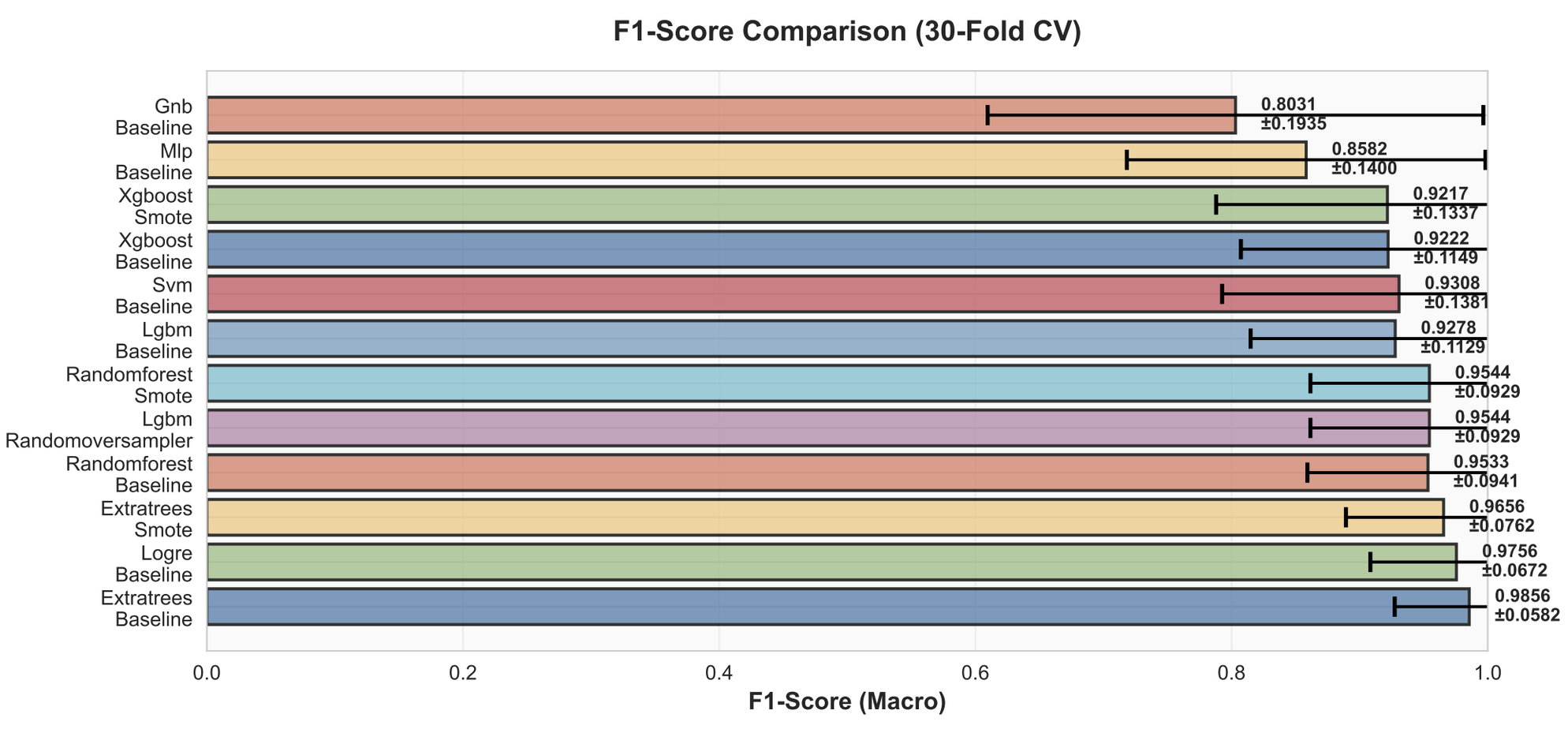
Dataset: Single Profiles
Date: December 24, 2025
Validation: 30-Fold Stratified Cross-Validation
Models Evaluated: 12

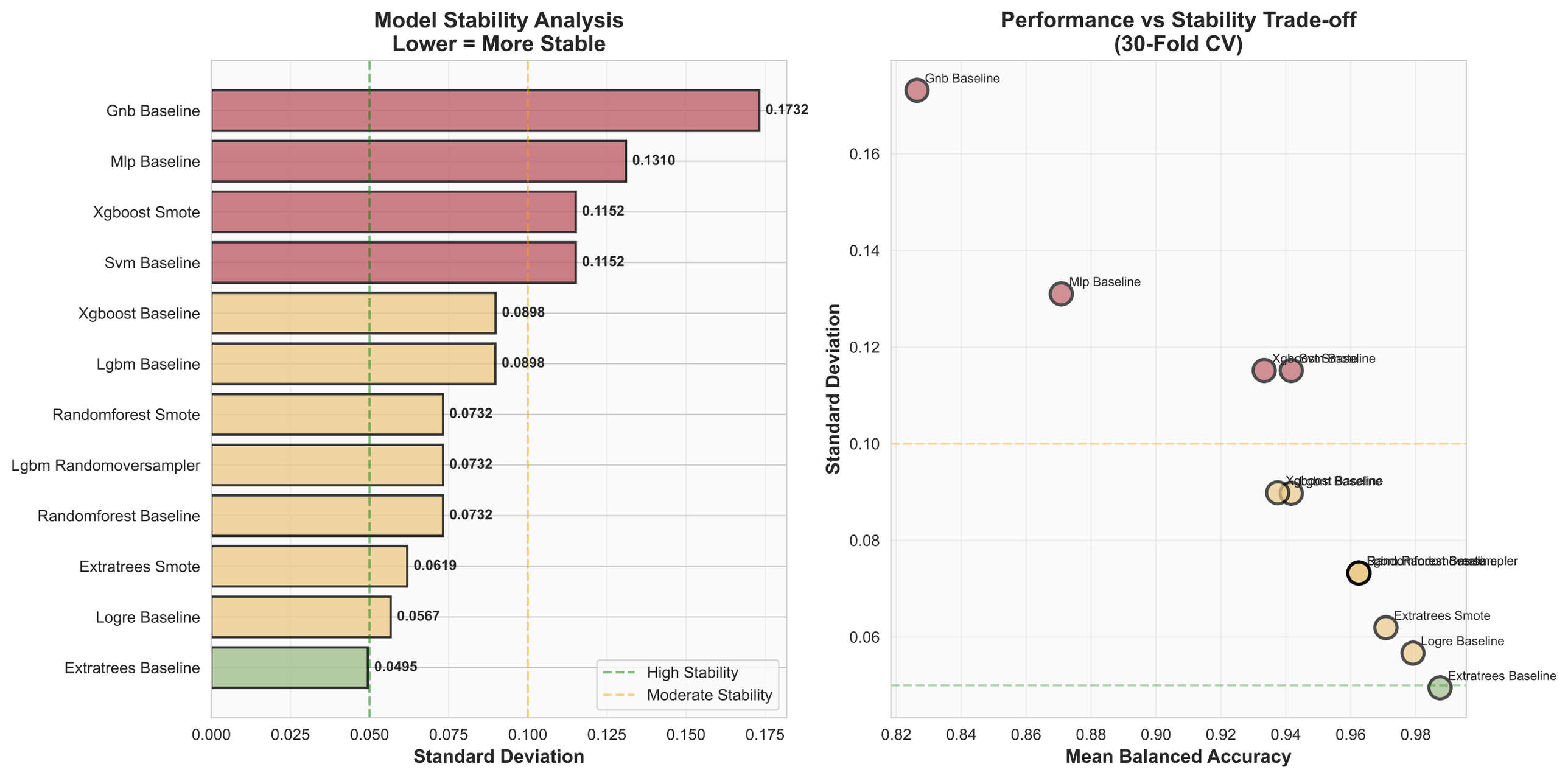
KEY FINDINGS

 Best Model: Extratrees Baseline
• Balanced Accuracy: 0.9875 ± 0.0495
• F1-Score (Macro): 0.9856
• Stability: High

 Validation Approach:
• 30-fold stratified cross-validation
• Ensures robust performance estimates
• Accounts for data variability
• Reduces overfitting bias


Single Dataset - Model Performance Comparison (30-Fold CV)
Target: 0.80
Gnb Baseline
0.8264 ±0.1732
Mlp Baseline
0.8708 ±0.1310
Xgboost Smote
0.9333 ±0.1152
Xgboost Baseline
0.9375 ±0.0898
Svm Baseline
0.9417 ±0.1152
Lgbm Baseline
0.9417 ±0.0898
Randomforest Smote
0.9625 ±0.0732
Lgbm Randomoversampler
0.9625 ±0.0732
Randomforest Baseline
0.9625 ±0.0732
Extratrees Smote
0.9708 ±0.0619
Logre Baseline
0.9792 ±0.0567
Extratrees Baseline
0.9875 ±0.0495
0.0
0.2
0.4
0.6
0.8
1.0
Balanced Accuracy


F1-Score Comparison (30-Fold CV)
Gnb Baseline
0.8031 ±0.1935
Mlp Baseline
0.8582 ±0.1400
Xgboost Smote
0.9217 ±0.1337
Xgboost Baseline
0.9222 ±0.1149
Svm Baseline
0.9308 ±0.1381
Lgbm Baseline
0.9278 ±0.1129
Randomforest Smote
0.9544 ±0.0929
Lgbm Randomoversampler
0.9544 ±0.0929
Randomforest Baseline
0.9533 ±0.0941
Extratrees Smote
0.9656 ±0.0762
Logre Baseline
0.9756 ±0.0672
Extratrees Baseline
0.9856 ±0.0582
0.0
0.2
0.4
0.6
0.8
1.0
F1-Score (Macro)


Model Stability Analysis
Lower = More Stable
Gnb Baseline
Mlp Baseline
Xgboost Smote
Svm Baseline
Xgboost Baseline
Lgbm Baseline
Randomforest Smote
Lgbm Randomoversampler
Randomforest Baseline
Extratrees Smote
Logre Baseline
Extratrees Baseline
0.1732
0.1310
0.1152
0.1152
0.0898
0.0898
0.0732
0.0732
0.0732
0.0619
0.0567
0.0495
High Stability
Moderate Stability
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
Standard Deviation
Performance vs Stability Trade-off
(30-Fold CV)
Gnb Baseline
Mlp Baseline
Extratrees Smote
Logre Baseline
Extratrees Baseline
0.16
0.14
0.12
0.10
0.08
0.06
Standard Deviation
0.82
0.84
0.86
0.88
0.90
0.92
0.94
0.96
0.98
Mean Balanced Accuracy

Single Dataset - Detailed Performance Metrics

Rank	Model	Bal Acc (Mean±Std)	F1 (Mean±Std)	Accuracy (Mean±Std)	Stability
1	Extratrees Baseline	0.9875±0.0495	0.9856±0.0582	0.9897±0.0387	High
2	Logre Baseline	0.9792±0.0567	0.9756±0.0672	0.9802±0.0507	Moderate
3	Extratrees Smote	0.9708±0.0619	0.9656±0.0762	0.9706±0.0589	Moderate
4	Randomforest Baseline	0.9625±0.0732	0.9533±0.0941	0.9611±0.0744	Moderate
5	Lgbm Randomoversampler	0.9625±0.0732	0.9544±0.0929	0.9611±0.0744	Moderate
6	Randomforest Smote	0.9625±0.0732	0.9544±0.0929	0.9611±0.0744	Moderate
7	Lgbm Baseline	0.9417±0.0898	0.9278±0.1129	0.9460±0.0801	Moderate
8	Svm Baseline	0.9417±0.1152	0.9308±0.1381	0.9500±0.0980	Variable
9	Xgboost Baseline	0.9375±0.0898	0.9222±0.1149	0.9405±0.0820	Moderate
10	Xgboost Smote	0.9333±0.1152	0.9217±0.1337	0.9413±0.0890	Variable
11	Mlp Baseline	0.8708±0.1310	0.8582±0.1400	0.8754±0.1207	Variable
12	Gnb Baseline	0.8264±0.1732	0.8031±0.1935	0.8381±0.1567	Variable